

A Technical Introduction to Machine translation

Section 1

1950s Yehoshua Bar-Hillel, began his research at MIT (1951). A Georgetown University MT research team, led by Professor Michael Zarechnak, followed (1951) with a public demonstration of its Georgetown-IBM experiment system in 1954. MT research programs popped up in Japan and Russia (1955), and the first MT conference was held in London (1956). David G. Hays "wrote about computer-assisted language processing as early as 1957" and "was project leader on computational linguistics at Rand from 1955 to 1968."

Section 2

The ideal deep approach would require the translation software to do all the research necessary for this kind of disambiguation on its own; but this would require a higher degree of AI than has yet been attained. A shallow approach which simply guessed at the sense of the ambiguous English phrase that Piron mentions (based, perhaps, on which kind of prisoner-of-war camp is more often mentioned in a given corpus) would have a reasonable chance of guessing wrong fairly often. A shallow approach that involves "ask the user about each ambiguity" would, by Piron's estimate, only automate about 25% of a professional translator's job, leaving the harder 75% still to be done by a human.

Section 3

A deep learning-based approach to MT, neural machine translation has made rapid progress in recent years. However, the current consensus is that the so-called human parity achieved is not real, being based wholly on limited domains, language pairs, and certain test benchmarks, i.e., it lacks statistical significance power. Translations by neural MT tools like DeepL Translator, which is thought to usually deliver the best machine translation results as of 2022, typically still need post-editing by a human. Instead of training specialized translation models on parallel datasets, one can also directly prompt generative large language models like GPT to translate a text. This approach is considered promising, but is still more resource-intensive than specialized translation models.

Section 4

1960–1975 Researchers continued to join the field as the Association for Machine Translation and Computational Linguistics was formed in the U.S. (1962) and the National Academy of Sciences formed the Automatic Language Processing Advisory Committee (ALPAC) to study MT (1964). Real progress was much slower, however, and

after the ALPAC report (1966), which found that the ten-year-long research had failed to fulfill expectations, funding was greatly reduced. According to a 1972 report by the Director of Defense Research and Engineering (DDR&E), the feasibility of large-scale MT was reestablished by the success of the Logos MT system in translating military manuals into Vietnamese during that conflict. The French Textile Institute also used MT to translate abstracts from and into French, English, German and Spanish (1970); Brigham Young University started a project to translate Mormon texts by automated translation (1971).

Section 5

Medicine Despite being labelled as an unworthy competitor to human translation in 1966 by the Automated Language Processing Advisory Committee put together by the United States government, the quality of machine translation has now been improved to such levels that its application in online collaboration and in the medical field are being investigated. The application of this technology in medical settings where human translators are absent is another topic of research, but difficulties arise due to the importance of accurate translations in medical diagnoses. Researchers caution that the use of machine translation in medicine could risk mistranslations that can be dangerous in critical situations. Machine translation can make it easier for doctors to communicate with their patients in day-to-day activities, but it is recommended to only use machine translation when there is no other alternative, and that translated medical texts should be reviewed by human translators for accuracy.

Section 6

Non-standard speech One of the major pitfalls of MT is its inability to translate non-standard language with the same accuracy as standard language. Heuristic or statistical based MT takes input from various sources in standard form of a language. Rule-based translation, by nature, does not include common non-standard usages. This causes errors in translation from a vernacular source or into colloquial language. Limitations on translation from casual speech present issues in the use of machine translation in mobile devices.

Section 7

Studies using human evaluation (e.g. by professional literary translators or human readers) have systematically identified various issues with the latest advanced MT outputs. Some quality evaluation studies have found that, in several languages, human translations outperform ChatGPT-produced translations in terminological accuracy and clarity of expression. Common issues include the translation of ambiguous parts whose correct translation requires common sense-like semantic language processing

or context. There can also be errors in the source texts, missing high-quality training data and the severity of frequency of several types of problems may not get reduced with techniques used to date, requiring some level of human active participation.

Section 8

Why does a translator need a whole workday to translate five pages, and not an hour or two? About 90% of an average text corresponds to these simple conditions. But unfortunately, there's the other 10%. It's that part that requires six [more] hours of work. There are ambiguities one has to resolve. For instance, the author of the source text, an Australian physician, cited the example of an epidemic which was declared during World War II in a "Japanese prisoners of war camp". Was he talking about an American camp with Japanese prisoners or a Japanese camp with American prisoners? The English has two senses. It's necessary therefore to do research, maybe to the extent of a phone call to Australia.

Section 9

Travel Machine translation applications have also been released for most mobile devices, including mobile telephones, pocket PCs, PDAs, etc. Due to their portability, such instruments have come to be designated as mobile translation tools enabling mobile business networking between partners speaking different languages, or facilitating both foreign language learning and unaccompanied traveling to foreign countries without the need of the intermediation of a human translator. For example, the Google Translate app allows foreigners to quickly translate text in their surrounding via augmented reality using the smartphone camera that overlays the translated text onto the text. It can also recognize speech and then translate it.